DSYLEXIA FRIENDLY READER

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PROJECT OVERVIEW

The Dyslexia-Friendly Reader is a web-based application designed to help people with dyslexia read text more comfortably and efficiently. Many users with dyslexia face difficulties such as crowded text, complex words, poor spacing, and visual stress while reading digital content.

This project improves readability by:

- Breaking large text into smaller readable sections
- Increasing spacing between lines
- Simplifying difficult words
- Allowing users to customize reading settings
- Maintaining reading history for future access

The application provides a clean and accessible interface that makes reading easier and reduces cognitive load for users.

PROBLEM STATMENT

People with dyslexia often struggle to read standard digital text because of:

- Poor text spacing
- Complex vocabulary
- Crowded paragraphs
- Lack of personalization in reading interfaces

Most websites and reading applications are not optimized for dyslexic readers, making it difficult for them to understand and process information efficiently.
The project aims to solve this problem by creating a customizable and accessible reading platform specially designed for dyslexic users.

OBJECTIVES

The main objectives of the project are:

To create a user-friendly reading platform for dyslexic users.

To improve readability through spacing and formatting adjustments.

To simplify difficult words into easier alternatives.

To allow users to customize their reading experience.

To provide login and history functionality for personalized usage.

To reduce reading stress and improve comprehension.

FEATURES

🔐 User Authentication
Signup and Login system
Session handling using localStorage
Logout functionality
📖 Dyslexia-Friendly Reading
Converts large text into readable chunks
Highlights separated sentences
Adjustable line spacing
🧠 Text Simplification
Replaces difficult words with simpler alternatives
Improves readability and comprehension
📊 Difficulty Checker
Analyzes text difficulty level
Displays:
Easy
Medium
Hard
⚙️ Custom Settings
Users can customize:
Font size
Letter spacing
Line spacing
Background color
📜 Reading History
Saves previously entered text
User-specific history storage
Delete individual history items
Clear all history
👤 Personalized Experience
Displays logged-in username
Separate history for each user

TECHNOLOGIES USED

Frontend

HTML5

CSS3

JavaScript

Storage

Browser LocalStorage

Development Environment

VS Code

XAMPP (for localhost testing)

ARCHITECTURE/WORLFLOW

Step 1: User Authentication
User creates an account using Signup page
Credentials are stored in localStorage
User logs in through Login page
↓
Step 2: Session Management
Application checks login status
Logged-in user session is maintained
↓
Step 3: Text Input
User enters or pastes text into the reader
↓
Step 4: Text Processing
The system:
Splits text into smaller readable sections
Simplifies difficult words
Applies dyslexia-friendly formatting
↓
Step 5: Difficulty Analysis
Word count is analyzed
Difficulty level is displayed
↓
Step 6: Save History
User text is saved in personalized history
↓
Step 7: Settings Customization
Users can adjust:
Font size
Line spacing
Letter spacing
Background color
↓
Step 8: Output Display
Processed text is displayed in an easier-to-read format.

CONCLUSION

The Dyslexia-Friendly Reader improves accessibility and inclusivity in digital reading by providing a personalized and user-friendly reading environment. The project demonstrates how web technologies can be used to support individuals with learning difficulties and improve their overall reading experience.

THANK YOU!